HONG HUANG

GitHub ◇ Google Scholar ◇ Personal Website

Phone: +86-17349764371 ◇ WeChat: Hong4Work ◇ Email: honghuang2000@outlook.com

RESEARCH INTERESTS

Efficient AI, Efficient Large Models, AI Infra, Model Compression & Acceleration, Edge Intelligence

EDUCATION/PROFESSIONAL EXPERIENCE

City University of Hong Kong — *Hong Kong, China; Sept. 2024 – Present*
Ph.D. in Computer Science — *Advised by Dr. Dapeng Wu*

City University of Hong Kong — *Hong Kong, China; Sept. 2023 – Aug. 2024*
Research Assistant in Computer Science — *Advised by Dr. Dapeng Wu*

University of Florida — *Gainesville, United States; Aug. 2021 – May 2023*
M.Sc. in Electrical and Computer Engineering — *Advised by Dr. Ruogu Fang and Dr. Dapeng Wu*

Shanghai Jiao Tong University — *Shanghai, China; Aug. 2017 – June 2021*
B.E. in Computer Science and Technology — *Advised by Dr. Jian Cao*

SELECTED PUBLICATIONS

(*Equal Contribution; †Corresponding Author)

Efficient LFMs (*Bar Menu*)

- **[Sherry ACL'26 (Oral)] Hong Huang**, Decheng Wu, Qiangqiang Hu, Guanghua Yu, Jinhai Yang, Jianchen Zhu, Xue Liu, Dapeng Wu. "Sherry: Hardware-Efficient 1.25-Bit Ternary Quantization via Fine-grained Sparsification." [PDF], [Code], [QbitAI (量子位)], [HY Translator Demo App (100K+ users)] [Tencent Micro-Innovation Award]
- **[Tequila ICLR'26] Hong Huang**, Decheng Wu, Rui Cen, Guanghua Yu, Zonghang Li, Kai Liu, Jianchen Zhu, Peng Chen, Xue Liu, Dapeng Wu. "Tequila: Trapping-free Ternary Quantization for Large Language Models." [PDF], [Code], [Tencent News]
- **[Aquavit SIGKDD'26] Hong Huang**, Jiaxun Ye, Jinhai Yang, Wenjiao Feng, Zonghang Li, Ning Chen. "Aquavit: Ascending Quantization for Communication-Efficient Vast-Scale Distributed Training." [PDF]
- **[Quaff ACL'25] Hong Huang**, Dapeng Wu. "Quaff: Quantized Parameter-Efficient Fine-Tuning under Outlier Spatial Stability Hypothesis." [PDF], [Code]

Efficient Federated Learning (*Fed Series*)

- **[FedRTS NeurIPS'25] Hong Huang**, Hai Yang, Yuan Chen, Jiaxun Ye, Dapeng Wu. "FedRTS: Federated Robust Pruning via Combinatorial Thompson Sampling." [PDF], [Code]
- **[FedFit ICML'26]** Meng Bi, **Hong Huang**†, Jinlong Song, Charles Wang, Chengming Hu, Xi Chen, Ting Yu, Xue Liu. "FedFit: Federated Dynamic Pruning via Fisher Information Scoring." [PDF]
- **[AquaFed TCC'26]** Juntao Hu*, **Hong Huang***, Kuan Liu, Huimin Lu, Bingyi Liu, Dapeng Wu. "AquaFed: Ascending Quantized Federated Learning on Heterogeneous Devices." IEEE Transactions on Cloud Computing. [PDF]
- **[FedMef CVPR'24] Hong Huang**, Weiming Zhuang, Chen Chen, and Lingjuan Lyu. "FedMef: Towards Memory-efficient Federated Dynamic Pruning." [PDF], [Code]
- **[FedTiny ICDCS'23] Hong Huang**, Lan Zhang, Chaoyue Sun, Ruogu Fang, Xiaoyong Yuan, and Dapeng Wu. "Distributed Pruning Towards Tiny Neural Networks in Federated Learning." [PDF], [Code]

Efficient MLSys (*.cpp Series*)

- **[Prima.cpp ICLR'26]** Zonghang Li, Tao Li, Wenjiao Feng, Rongxing Xiao, Jianshu She, **Hong Huang**, Mohsen Guizani, Hongfang Yu, Qirong Ho, Wei Xiang, Steve Liu. "Prima.cpp: Fast 30-70B LLM Inference on Heterogeneous and Low-Resource Home Clusters." [PDF], [Code]

INTERNSHIP EXPERIENCE

Tencent *Shenzhen, China; Aug. 2025 – Present*
Research Intern (Qingyun Top 青云计划), AI Infra *Mentored by Jianchen Zhu*

- Focused on ultra-low-bit quantization techniques. Proposed **Tequila** (1.58-bit) and **Sherry** (1.25-bit), achieving breakthroughs in ternary quantization with near-lossless performance versus FP16 on ARC metrics. Tequila and Sherry were accepted at **ICLR'26** and **ACL'26 (Oral)**, respectively; Sherry received the **Tencent Micro-Innovation Award 2026**.
- Developed the HY Translator demo app, a fully offline translator using only 700 MB peak RAM, reaching 100K+ downloads within one month. Built the Hy-MT2 model series of on-device translation models supporting 33 languages, ranking **Top-1** and **Top-3** on the Hugging Face Weekly Trending list.

SONY AI *Tokyo, Japan; Mar. 2023 – Aug. 2023*
Research Intern, Privacy-Preserving Machine Learning (PPML) Team *Mentored by Dr. Lingjuan Lyu*

- Developed FedMef, a novel memory-efficient federated dynamic pruning framework.
- Achieved 28.5% memory savings while improving accuracy by 2%; published at **CVPR'24** [PDF].

Meta *Menlo Park, United States; Mar. 2022 – Dec. 2022*
Research Assistant, Video Infrastructure Group *Mentored by Dr. Zhijun Lei*

- Developed TMAP, a CNN-based texture- and motion-aware in-loop filter for AV1.
- Achieved a 4.32% BD-rate reduction and 3.79% VMAF gain; published in **JVCIR** [PDF].

LEADERSHIP

- Lead the Efficient & Reliable AI (ERA) Research Lab, a group of 15+ junior Ph.D. and M.Sc. students focused on efficient and reliable AI; coordinating research that yielded 6 papers accepted to top-tier conferences and journals in 2025–2026 (NeurIPS'25, IEEE TCC'26, ICLR'26, ACL'26, ICML'26, SIGKDD'26).

AWARD/SCHOLARSHIP/FELLOWSHIP

Tencent Micro-Innovation Award — Tencent 2026
DAAD AINet Fellowship (Postdoc-NeT-AI) — German Academic Exchange Service 2025
Research Tuition Scholarship — City University of Hong Kong 2025
Graduate School Fellowship — University of Florida 2021–2023
Zhiyuan Honor Scholarship — Shanghai Jiao Tong University 2017–2021